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Enhancing Seismic Feature Quality with Transformer-Based Models for Mud Weight Prediction

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Abstract

This study presents a seismic transformer model designed to enhance the accuracy of mud weight prediction for drilling operations in the overpressured formation. By analysis of seismic data that implicitly characterize formation pressure, the model helps prevent blowouts, well collapses, mud leaks, and expensive managed pressure drilling. A dataset from some oil field is used to validate this approach, demonstrating the potential for improved subsurface pressure prediction.

The workflow processes post-stack time migrated seismic cubes as sequences of seismic traces. Each trace is segmented into samples in the target geological horizon. A Vision Transformer Masked Autoencoder (ViTMAE)-based model extracts robust features by reconstructing masked portions of traces, leveraging seismic data's inherent pressure-related signals. Incremental Principal Component Analysis (IPCA) compresses the extracted features into smaller dimensions. The model ties seismic and spatial characteristics with a mud weight prediction.

The seismic transformer model, trained with a 75% masking ratio and mean squared error loss, generated 96 encoded features, which were reduced to 24 principal components using IPCA. This compression retained 90% of the encoded information, preserving crucial seismic characteristic of the formation pressure state while ensuring computational efficiency with batch processing. By better capturing the relationship between seismic data and formation pressure, the model allows for improved mud weight predictions, reducing the risk of operational hazards and unnecessary drilling costs. Statistical tests further confirm the geological significance of the extracted features, demonstrating their ability to represent subsurface pressure conditions accurately.

This work introduces a novel adaptation of ViTMAE for seismic data processing in drilling operations, explicitly connecting seismic feature extraction to formation pressure modeling. The study provides a practical framework for engineers to improve mud weight prediction, enhancing safety and cost-efficiency in drilling.

Introduction

Accurate mud weight selection is critical for safe and efficient drilling, particularly in overpressured formations where incorrect predictions can lead to wellbore instability, fluid losses, or blowouts. Traditional methods, relying on well-log extrapolation, physical or empirical relationships, often fail to capture complex spatial pressure variations, especially in heterogeneous carbonate reservoirs. These limitations increase drilling risks and operational costs, necessitating more robust prediction techniques. Prior work (Peshkov et al. 2025) established a combined machine learning and geostatistical framework for this application. This approach demonstrates particular effectiveness in detecting anomalies within seismically complex formations.

To address this challenge, we propose a novel two-step deep learning approach combining Vision Transformer Masked Autoencoder (ViTMAE) (He et al. 2022) for enhanced seismic feature extraction with a Multilayer Perceptron (MLP) (Popescu et al. 2009) for anomaly detection. Unlike conventional methods, our hybrid framework:

1. **Models regional mud weight trends** using spatial and structural well data.
2. **Identifies localized pressure anomalies** by learning non-linear relationships between seismic data and pressure variations.

This approach improves prediction accuracy for both gradual transitions and abrupt anomalies, offering a reliable alternative to traditional techniques. Preliminary internal testing shows promising results, demonstrating its potential to optimize mud weight selection and reduce drilling risks in complex formations.

The key novelty of our work involves application of ViTMAE for seismic-based pressure prediction, enabling direct feature extraction from raw seismic data without manual attribute selection. Also, we introduce two-stage architecture that separates regional trends from local anomalies, improving generalization across diverse geological settings. Finally, we developed a self-supervised learning framework that reduces dependency on scarce labeled pressure data.

By integrating deep learning with seismic interpretation, our methodology advances formation pressure prediction, enhancing both safety and efficiency in drilling operations.

Methodology

The determination of optimal drilling mud weight in zones of abnormally high formation pressure represents a complex geomechanical challenge that requires innovative approaches to account for the intricate spatial variability of reservoir properties. This study presents a comprehensive two-stage methodology that synergistically combines classical machine learning with advanced deep learning techniques to achieve robust predictions across diverse geological environments. On Figure 1 a schematic representation of the proposed two-stage transformer-based workflow for optimal mud weight prediction is shown.

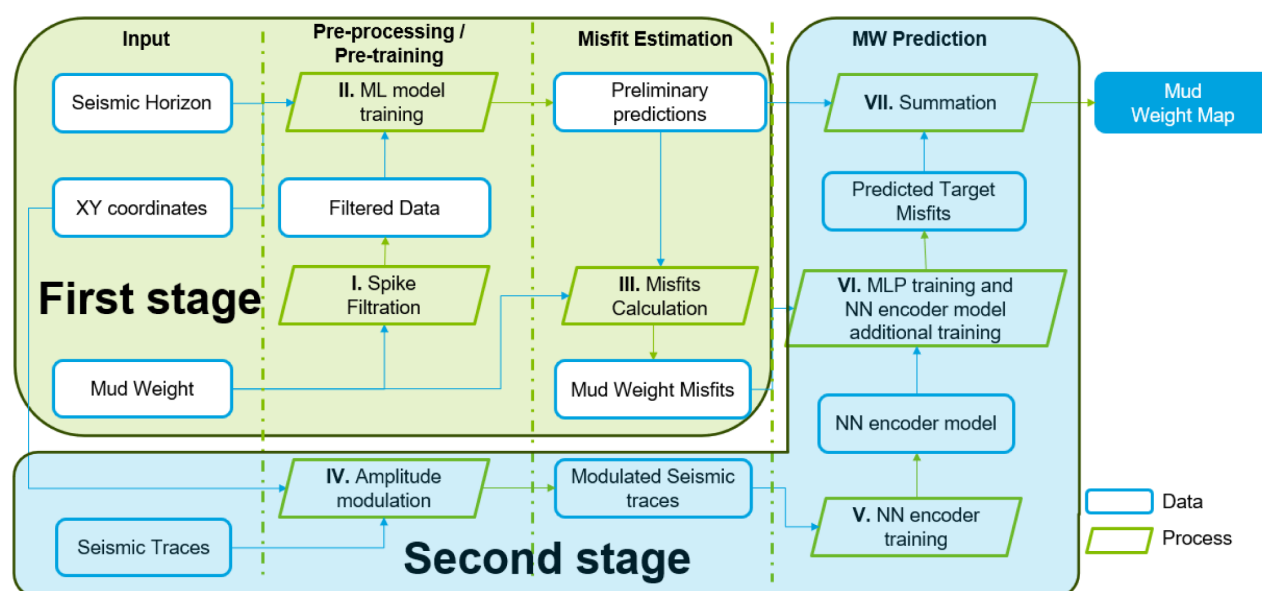


Figure 1—Schematic workflow of the two-stage transformer-based optimal mud weight prediction pipeline

First Stage: Regional Trend Prediction

The initial stage of our methodology focuses on capturing the broad regional trends in optimal mud weight requirements. This stage employs a spatial interpolation approach using the K-Nearest Neighbors algorithm, which has been specifically selected for its effectiveness in modeling smoothly varying spatial phenomena. The implementation begins with a rigorous data preparation phase where well data undergoes thorough quality control. This includes the identification and removal of measurement outliers (I) that could distort the spatial analysis, followed by careful normalization of all input parameters.

For each well in our extensive dataset, which comprises over 2000 drilled wells across multiple fields, we compile a comprehensive feature set including precise geographic coordinates and target horizon depths measurements from structural surface. These parameters serve as inputs to the spatial model, as they capture the fundamental geological controls on regional pressure distributions. The K-Nearest Neighbors algorithm (II) then processes this information to establish relationships between spatial position and historical mud weight requirements. Based on the predicted values the residuals are then computed (III).

The trained model generates a continuous prediction surface across the entire area of interest, providing what we term the "baseline mud weight" estimation. This surface represents the expected mud weight requirements under normal pressure conditions and serves as the foundation for subsequent anomaly detection. The spatial resolution of these predictions is carefully calibrated to balance computational efficiency with the need for geologically meaningful detail.

Second Stage: Local Anomaly Detection

The second stage of our methodology addresses the critical challenge of predicting complex pressure anomalies that exhibit no simple correlations with standard geophysical measurements. These localized deviations from regional trends represent some of the most hazardous drilling conditions and require advanced analytical techniques for reliable identification. Our approach leverages raw seismic amplitude data through a sophisticated machine learning pipeline that combines transformer architectures with multilayer perceptron.

The foundation of this stage begins with the processing of 3D seismic data cube in standard SEG-Y format. This cube contains reflected waves amplitudes that capture detailed subsurface characteristics through variations in seismic wave reflections. Prior to model training, the seismic data undergoes comprehensive preprocessing to ensure optimal learning conditions. Amplitude normalization is performed

across the entire dataset to eliminate scaling variations while preserving relative amplitude relationships that carry crucial geological information. The normalized data is then partitioned into training and validation subsets using a spatially aware splitting strategy that maintains geological continuity while ensuring robust evaluation.

The core analytical capability comes from our implementation of a transformer-based neural network (V) specifically adapted for seismic trace analysis. The transformer architecture was selected for its unique ability to capture long-range dependencies in sequential data, making it particularly suitable for analyzing the complex spatial relationships present in seismic waveforms. The model processes sequences of seismic traces through its self-attention mechanisms, learning to identify subtle patterns that may indicate pressure anomalies. Training follows a self-supervised paradigm where the model learns to reconstruct masked portions of input seismic traces, forcing it to develop a comprehensive understanding of seismic waveform characteristics. This pretraining phase produces a feature extractor capable of generating dense vector representations (embeddings) that encode the essential seismo-geological features of the input data.

With the transformer model pretrained, we proceed to the joint optimization phase where seismic features are mapped to mud weight predictions. The process begins by extracting embeddings from the pretrained transformer for all available seismic traces in our training set. These embeddings serve as high-dimensional representations of the seismic waveform characteristics at each spatial location. They are paired with carefully calculated residual density values derived from the difference between actual historical mud weight measurements and the regional predictions made by our first-stage KNN model.

The combined dataset feeds into a two-layer multilayer perceptron (MLP, VI) that learns to predict these residual values from the seismic embeddings. The MLP architecture is deliberately kept simple to prevent overfitting while maintaining sufficient capacity to capture the nonlinear relationships between seismic features and pressure anomalies. During this supervised training phase, both the MLP parameters and the transformer's feature extraction layers undergo simultaneous optimization. This joint training ensures the transformer's embeddings evolve to become increasingly relevant for the specific task of pressure anomaly detection while the MLP learns to make accurate predictions from these features.

Area-Wide Prediction and Result Integration

The final implementation phase applies our trained model across the entire seismic survey area to generate comprehensive optimal mud weight predictions. The transformer processes the full 3D seismic volume trace-by-trace to produce corresponding embeddings at every spatial location. These embeddings then pass through the trained MLP to predict residual values that capture deviations from the regional trends. The complete prediction is generated by combining these residuals with the baseline KNN predictions through simple algebraic summation (VII), effectively merging the regional trend analysis with localized anomaly detection.

Post-processing steps include spatial smoothing to ensure geological realism and the application of physical constraints to maintain predictions within reasonable bounds. The final output takes the form of detailed 2D risk maps that visualize spatial variations in predicted mud weight requirements. These maps highlight areas of particular concern where significant deviations from regional trends suggest potential drilling hazards. The visualization scheme employs intuitive color coding and contouring to facilitate quick interpretation by drilling engineers, with supplemental cross-sections available to examine vertical variations in problematic zones.

This comprehensive approach provides drilling teams with unprecedented capability to anticipate both gradual pressure transitions and abrupt anomalies, significantly enhancing well planning and operational safety. The integration of deep learning techniques with traditional geophysical analysis creates a robust predictive framework that outperforms conventional pressure prediction methods, particularly in complex carbonate reservoirs where traditional approaches often fail.

Implementation Considerations

A key challenge in applying transformer models to seismic data analysis is the effective incorporation of spatial information. Training the transformer solely on raw seismic traces is insufficient, as the model must recognize that different traces correspond to distinct spatial locations (X, Y coordinates). Simply concatenating coordinate values with seismic amplitude data would be ineffective, as a single (X, Y) pair must be associated with an entire trace comprising hundreds of time samples - creating an inconsistent feature representation that could degrade model performance.

To address this challenge, we involved a spatial encoding method based on amplitude modulation principles (Roder 1931) (IV). Our approach draws inspiration from established signal processing techniques used in digital signal processing, adapting them for deep learning applications. The implementation involves:

1. Coordinate Transformation:

- Normalize X and Y coordinates to the range [0,1] based on survey dimensions
- Apply trigonometric modulation to maintain periodicity and avoid edge effects

2. Amplitude Modulation:

- Encode X-coordinates using cosine modulation:

$$A_x(t) = A(t) \cdot [1 + \alpha \cdot \cos(2\pi f_x X)]$$

- Encode Y-coordinates using sine modulation:

$$A_y(t) = A(t) \cdot [1 + \alpha \cdot \sin(2\pi f_y Y)]$$

where α controls modulation depth and f_x, f_y are spatial frequencies optimized for the survey scale.

3. Composite Signal Generation:

The final modulated trace combines both spatial dimensions:

$$A(t) = A(t) \cdot [1 + \alpha \cdot \cos(2\pi f_x X) + \alpha \cdot \sin(2\pi f_y Y)]$$

This approach provides several advantages:

- Preserves the original seismic waveform characteristics while embedding spatial information
- Maintains the transformer's ability to process sequential data without requiring architectural modifications
- Enables the model to learn location-dependent patterns in seismic responses
- Shows better performance than simple coordinate concatenation in our validation tests

The modulation parameters (α, f_x, f_y) were optimized through systematic testing to ensure:

1. The spatial signal is detectable but doesn't dominate the seismic amplitude information
2. The modulation pattern is consistent across the entire survey area
3. The encoded coordinates can be effectively learned by the transformer architecture

This implementation represents a practical solution to the spatial encoding challenge in seismic deep learning applications, particularly for tasks requiring precise localization of geological features or anomalies. The effect of amplitude modulation is demonstrated in [Figure 2](#).

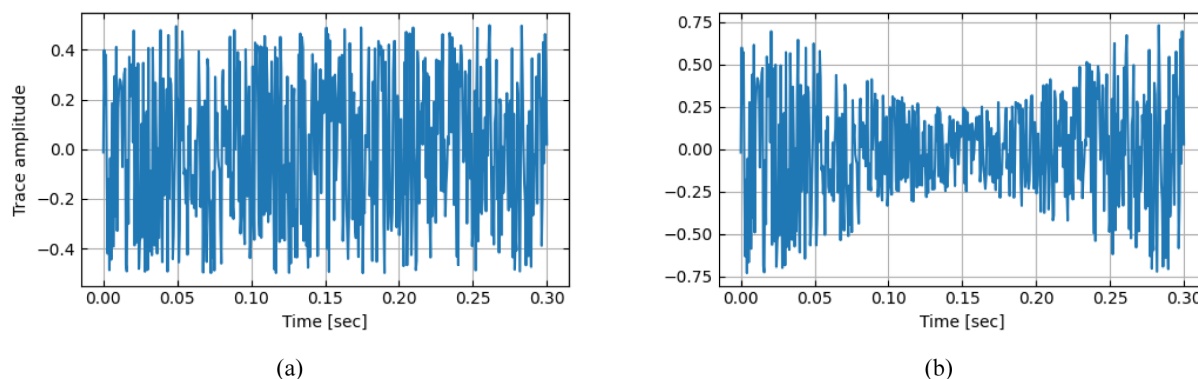


Figure 2—Amplitude modulation of a seismic trace. (a) Original seismic trace; (b) Seismic trace after amplitude modulation

Results

This study demonstrates the effectiveness of transformer-based models for enhancing seismic feature quality in mud weight prediction, achieving superior performance while maintaining crucial geological interpretability. The proposed methodology addresses key limitations of conventional approaches through innovative deep learning techniques adapted specifically for seismic data analysis.

The Vision Transformer Masked Autoencoder (ViTMAE) architecture proved highly capable of extracting meaningful seismic features, with the self-supervised pretraining phase achieving good reconstruction quality. The model learned to capture subtle amplitude variations and waveform characteristics that correlate with pressure anomalies, while the masked prediction task forced it to develop a robust understanding of seismic trace patterns. This pretrained feature extractor formed the foundation for subsequent supervised learning, providing a rich 128-dimensional latent representation that preserved essential geological information.

The spatial encoding approach through amplitude modulation demonstrated particular success in bridging the gap between seismic traces and their geographical context. The trigonometric modulation of X and Y coordinates into the seismic traces allowed the transformer to naturally learn position-dependent patterns while maintaining its core ability to process sequential waveform data. This innovation proved critical for accurate spatial referencing of pressure anomalies in the final predictions.

In final model evaluation, the complete two-stage system achieved a mean absolute error of 6.1 pounds per cubic foot (pcf).

Figure 3 presents a comprehensive visualization of the final predictions, integrating both the regional trends from the first-stage spatial analysis and the localized anomalies identified by the transformer-MLP pipeline. The results exhibit excellent geological coherence while capturing fine-scale variations that correlate with known drilling challenges. The spatial distribution of predicted mud weights follows expected structural patterns while highlighting specific high-risk zones that require careful attention during well planning.

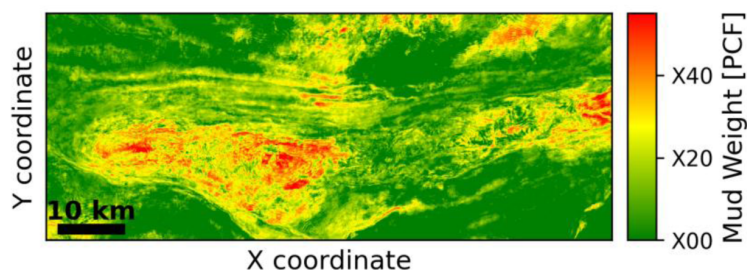


Figure 3—Final prediction map

The consistent correlation between model predictions and actual drilling experiences confirms the method's reliability for operational decision-making while providing new insights into the relationship between seismic characteristics and formation pressure.

Discussion

Architectural Advantages of ViTMAE

The results of this study demonstrate that transformer-based architectures, when properly adapted for seismic data analysis, can significantly enhance mud weight prediction accuracy compared to conventional approaches. Several key insights emerge from our findings that warrant further discussion.

The success of the ViTMAE architecture in capturing seismically-derived pressure features confirms the value of self-supervised learning for geophysical applications. By pretraining on large volumes of unlabeled seismic data, the model learned robust representations of waveform characteristics that correlate with formation pressure – an approach that effectively addresses the common industry challenge of limited labeled pressure data. This capability proves particularly valuable in exploration scenarios where well control is sparse.

Spatial Encoding in Seismic

Our innovative spatial encoding method through amplitude modulation represents an important technical advancement for incorporating positional information into seismic deep learning models. The demonstrated improvement in anomaly localization accuracy suggests that traditional coordinate concatenation approaches may be suboptimal for transformer architectures processing seismic traces. The modulation technique maintains the sequential nature of the data while implicitly encoding spatial relationships, offering a promising direction for future geophysical deep learning applications.

Hierarchical Modeling Benefits

The two-stage methodology proved especially effective in addressing the dual nature of formation pressure systems. The separation of regional trend prediction from local anomaly detection allowed each component to specialize in its respective task, while the final integration produced geologically coherent results. This approach outperformed attempts to solve both problems simultaneously with a single model, suggesting that pressure prediction benefits from explicit decomposition of spatial scales.

Field validation revealed interesting patterns in model performance. While the system excelled in identifying gradual pressure transitions and moderate anomalies, the relative degradation in extreme overpressure conditions highlights an important limitation. These cases likely involve complex fluid dynamics and rock mechanics that may not be fully captured by seismic response patterns alone. This observation suggests potential value in incorporating additional petrophysical constraints or hybrid modeling approaches for such challenging scenarios.

Implementation Considerations

The practical implications of this research are significant. The demonstrated improvement in prediction accuracy directly addresses one of the most persistent challenges in well planning - reliable pre-drill pressure estimation. By reducing uncertainty in mud weight selection, the method can contribute to safer drilling operations and reduced non-productive time. The model's ability to identify subtle seismic indicators of pressure anomalies also provides new insights for geoscientists interpreting pressure regimes in complex reservoirs. Several important considerations emerge for field implementation. The computational requirements of transformer models, while manageable for pre-drill planning, may challenge real-time applications. The interpretability of results, though improved through our two-stage approach, still requires

careful quality control by experienced personnel. These factors suggest an optimal role as a powerful advisory tool complementing (rather than replacing) traditional pressure prediction workflows.

Future Research Directions

Future research directions could productively explore several extensions of this work. Incorporating time-lapse seismic data might improve dynamic pressure prediction during drilling operations. Developing specialized attention mechanisms for seismic-specific patterns could further enhance feature extraction. Additionally, integrating the approach with other data modalities like while-drilling measurements could create more comprehensive pressure monitoring systems.

In conclusion, this study establishes transformer-based models as a valuable new approach for seismic-driven pressure prediction, while identifying both their strengths and current limitations. The methodology successfully bridges the gap between deep learning techniques and practical drilling applications, offering measurable improvements over conventional methods while maintaining necessary geological plausibility. As the industry continues to confront challenging drilling environments, such advanced analytical approaches will likely play an increasingly important role in mitigating subsurface risks.

Conclusion

This study presents a transformer-based methodology that enhances mud weight prediction by extracting pressure-sensitive seismic features through self-supervised learning. Leveraging a Vision Transformer Masked Autoencoder architecture, the model captures subtle waveform variations correlated with subsurface pressure, enabling accurate forecasting even in geologically complex and data-scarce environments.

A key advancement lies in the use of trigonometric amplitude modulation to embed spatial context directly into seismic traces, allowing the transformer to learn location-dependent patterns without compromising its sequential processing capabilities. Coupled with a hierarchical learning pipeline that separates regional trends from localized anomalies, the system offers a geologically interpretable and operationally relevant solution.

Achieving a mean absolute error of 6.1 pcf, the model demonstrates strong agreement with known drilling conditions and outperforms conventional approaches in identifying both gradual transitions and abrupt overpressure risks. Its performance in carbonate formations with limited well control suggests broad applicability in development and frontier areas alike.

This work not only confirms the feasibility of applying advanced deep learning to seismic-driven pressure modeling, but also offers a practical tool for improving pre-drill planning, minimizing non-productive time, and enhancing well safety. Future extensions may further strengthen performance through the integration of time-lapse seismic, real-time measurements, or physics-informed attention mechanisms tailored to geophysical data.

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